

Qualification Pack



Solar Panel Installation and Maintenance Technician (O-Level 'Solar Power Electronics')

QP Code: NIE/ELE/Q0902

Version: 1.0

NSQF Level: 4

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NIE/ELE/Q0902: Solar Panel Installation and Maintenance Technician (O-Level 'Solar Power Electronics')

Brief Job Description

Nature: The Certificate course is targeted for creating qualified professionals in the field of Solar Panel Installation. Qualification has been developed in consultation with industry experts in the domain, aiming at Empowering the future workforce with necessary skills for employment and entrepreneur development of the qualifier. **Purpose:** The objective of this O Level (Solar Panel Installation and Maintenance Technician) course is to equip candidates with the knowledge and practical skills needed to safely and effectively install and maintain photovoltaic systems for renewable energy generation.

Personal Attributes

This course demands technical expertise in solar technology, hands-on installation skills, and meticulous attention to detail. Learners must be safety-conscious, physically fit, and adept at troubleshooting and innovating for efficiency. Adaptability, continuous learning, and effective communication are key for success.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)
2. [NIE/ELE/N0910: Solar Technology Fundamentals](#)
3. [NIE/ELE/N0911: Electric Fundamentals for Solar System](#)
4. [NIE/ELE/N0912: Solar Panel Installation](#)
5. [NIE/ELE/N0814: Safety, Maintenance, and Repairs of Solar Panel](#)
6. [NIE/ELE/N0922: Implementation of Solar Power Electronics Project](#)

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	Renewable Energy Electronics
Occupation	Electronics Assembly, Electronics Equipment Services

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Country	India
NSQF Level	4
Credits	20
Aligned to NCO/ISCO/ISIC Code	7421.1401
Minimum Educational Qualification & Experience	12th Class (Completed 12th or equivalent.) with NA of experience OR Pursuing 2nd year of 3-year regular Diploma (after 10th) (in Electronics and Commutation Engineering/ Electrical Engineering/CS/IT and allied branches) with NA of experience OR Previous relevant Qualification of NSQF Level (3) with 3 Years of experience in Electronics and Commutation Engineering/ Electrical Engineering/CS/IT and allied branches OR Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience in Electronics and Commutation Engineering/ Electrical Engineering/CS/IT and allied branches
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	30/05/2027
NSQF Approval Date	30/05/2024
Version	1.0
Reference code on NQR	QG-04-EH-02597-2024-V1-NIELIT
NQR Version	1

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

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NIE/ELE/N0910: Solar Technology Fundamentals

Description

Solar Technology Fundamentals equips students with a thorough understanding of solar energy, covering its historical development, environmental benefits, and its critical role in both global and Indian energy landscapes. Students will learn the fundamental principles of solar radiation, including measurement techniques and site assessment methods, to ensure optimal solar panel placement. The course will also provide expertise in various solar photovoltaic (PV) systems, including different types of panels, cell technologies, and system configurations such as rooftop, ground mount, and floating panels. Through a combination of theoretical knowledge and practical skills, students will be prepared to design and implement effective solar PV systems, select the right panels for different applications, and address factors that impact efficiency and longevity.

Scope

The scope covers the following :

- Solar Technology Fundamentals module provides a comprehensive understanding of renewable energy sources, the historical development of solar technology, and its environmental significance. It encompasses global and Indian energy scenarios and policies. The module covers the physics of solar radiation, measurement techniques, site assessment tools, and shade analysis for optimal panel placement. It also includes various solar PV technologies, such as different module types, cell technologies, and system configurations like rooftop, ground mount, and floating panels. Practical skills in designing efficient solar PV systems and selecting appropriate panels for specific applications are emphasized.

Elements and Performance Criteria

Introduction to Solar Energy

To be competent, the user/individual on the job must be able to:

- PC1.** Identify various renewable energy sources, outline the historical development of solar technology, and explain the role of solar energy in environmental sustainability.
- PC2.** Describe the global and Indian energy scenarios, and understand relevant energy policies that promote solar energy usage.

Solar Radiation and Site Assessment

To be competent, the user/individual on the job must be able to:

- PC3.** Explain the fundamental physics of solar radiation, differentiate between global beam and diffuse radiation, and use appropriate tools and methods to measure solar radiation accurately.
- PC4.** Apply site assessment techniques and tools to evaluate solar energy potential and perform shade analysis to determine the optimal placement of solar panels for maximum efficiency.

Solar Photovoltaic (PV) Systems Overview

To be competent, the user/individual on the job must be able to:

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- PC5.** Describe the fundamentals of solar PV systems, including module types, cell technologies, and concentrators, and identify various system types (rooftop, ground mount, floating panels, solar parks), while evaluating solar resources, site conditions, and environmental factors that impact system performance.

Solar Panel Types and Selection

To be competent, the user/individual on the job must be able to:

- PC6.** Identify various types of solar panels, understand their materials and manufacturing processes, evaluate factors affecting efficiency and lifespan, and apply guidelines to select the appropriate panels for specific applications.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The various renewable energy sources, their characteristics, and their roles in environmental sustainability.
- KU2.** The historical development of solar technology and its impact on current solar energy practices.
- KU3.** The physics of solar radiation, including global beam and diffuse radiation, and methods for measuring and analyzing solar radiation.
- KU4.** Different types of solar PV systems, including rooftop, ground mount, and floating panels, and their applications.
- KU5.** Various solar panel types, materials, manufacturing processes, factors affecting panel efficiency and lifespan, and guidelines for selecting appropriate panels for specific applications.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Effectively analyze and interpret data related to solar radiation and site assessments to determine optimal panel placement and system performance.
- GS2.** Design and implement solar PV systems, including selecting appropriate panel types, understanding different cell technologies, and adapting to various environmental conditions.
- GS3.** Evaluate and compare the efficiency and lifespan of different solar panels, considering materials and manufacturing processes.
- GS4.** Stay updated with global and local energy policies and scenarios, integrating this knowledge into solar technology applications and practices.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Solar Energy</i>	50	22	-	-
PC1. Identify various renewable energy sources, outline the historical development of solar technology, and explain the role of solar energy in environmental sustainability.	-	-	-	-
PC2. Describe the global and Indian energy scenarios, and understand relevant energy policies that promote solar energy usage.	-	-	-	-
<i>Solar Radiation and Site Assessment</i>	-	-	-	-
PC3. Explain the fundamental physics of solar radiation, differentiate between global beam and diffuse radiation, and use appropriate tools and methods to measure solar radiation accurately.	-	-	-	-
PC4. Apply site assessment techniques and tools to evaluate solar energy potential and perform shade analysis to determine the optimal placement of solar panels for maximum efficiency.	-	-	-	-
<i>Solar Photovoltaic (PV) Systems Overview</i>	-	-	-	-
PC5. Describe the fundamentals of solar PV systems, including module types, cell technologies, and concentrators, and identify various system types (rooftop, ground mount, floating panels, solar parks), while evaluating solar resources, site conditions, and environmental factors that impact system performance.	-	-	-	-
<i>Solar Panel Types and Selection</i>	-	-	-	-
PC6. Identify various types of solar panels, understand their materials and manufacturing processes, evaluate factors affecting efficiency and lifespan, and apply guidelines to select the appropriate panels for specific applications.	-	-	-	-
NOS Total	50	22	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0910
NOS Name	Solar Technology Fundamentals
Sector	Electronics
Sub-Sector	
Occupation	Electronics Assembly, Electronics Equipment Services
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

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NIE/ELE/N0911: Electric Fundamentals for Solar System

Description

Electric Fundamentals for Solar System equips students with a comprehensive understanding of basic electrical principles, circuit components, and fundamental electrical laws. Students will develop practical skills in connecting PV modules and integrating inverters and batteries, while also applying knowledge of magnetism and electromagnetic induction to design and troubleshoot solar power systems.

Scope

The scope covers the following :

- Electric Fundamentals for Solar System covers basic electrical principles, circuit components, and fundamental laws essential for solar installations. It includes practical skills for connecting PV modules, configuring circuits, and integrating inverters and batteries, along with applying magnetism and electromagnetic induction for designing and troubleshooting solar power systems.

Elements and Performance Criteria

Electrical Basics for Solar Installers

To be competent, the user/individual on the job must be able to:

PC1. Demonstrate a comprehensive understanding of electrical principles, circuit construction, and the application of magnetism, electromagnetism, and electronic components in solar power systems.

Connection of PV Module:

To be competent, the user/individual on the job must be able to:

PC2. Demonstrate proficiency in configuring PV modules in series, parallel, and combined series-parallel circuits, ensuring optimal electrical performance and efficiency.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Basic electrical principles including electron and conventional flow, current effects, circuit components, and fundamental electrical laws and definitions.
- KU2.** The construction and analysis of electrical circuits involving conductors, insulators, semiconductors, resistors, and circuit networks.
- KU3.** The principles of magnetism, electromagnetism, electromagnetic induction, and mutual induction, and their application in designing and troubleshooting solar power systems.
- KU4.** The configuration of PV modules in series, parallel, and combined series-parallel circuits, and the integration of cell connections, arrays, inverters, and batteries in solar installations.

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** Apply basic electrical principles to analyze and troubleshoot electrical circuits and components effectively.
- GS2.** Configure and connect PV modules in series, parallel, and combined circuits, ensuring proper integration with inverters and batteries.
- GS3.** Utilize knowledge of magnetism, electromagnetism, and electromagnetic induction to design, install, and maintain solar power systems.
- GS4.** Interpret and apply fundamental electrical laws and definitions in practical solar system installations and troubleshooting scenarios.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Electrical Basics for Solar Installers</i>	50	23	-	-
PC1. Demonstrate a comprehensive understanding of electrical principles, circuit construction, and the application of magnetism, electromagnetism, and electronic components in solar power systems.	-	-	-	-
<i>Connection of PV Module:</i>	-	-	-	-
PC2. Demonstrate proficiency in configuring PV modules in series, parallel, and combined series-parallel circuits, ensuring optimal electrical performance and efficiency.	-	-	-	-
NOS Total	50	23	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0911
NOS Name	Electric Fundamentals for Solar System
Sector	Electronics
Sub-Sector	
Occupation	Electronics Assembly, Electronics Equipment Services
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

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NIE/ELE/N0912: Solar Panel Installation

Description

The Solar Panel Installation module equips students with the skills to design and configure solar PV systems, including selecting mounting and racking systems, solar tracking options, and energy storage components. Students will gain technical expertise in sizing PV cells, batteries, DC-DC converters, charge controllers, and inverters (both grid-connected and off-grid) to ensure optimal system performance. The module also covers essential safety practices and compliance standards, including structural integrity, electrical safety, and proper cable and earthing requirements for effective and reliable solar energy installations.

Scope

The scope covers the following :

- Solar Panel Installation provides a comprehensive understanding of designing and configuring solar PV systems, including selecting appropriate mounting and racking systems, solar tracking options, and sizing key components like PV cells, batteries, and inverters. It also covers technical expertise in cable and earthing requirements, along with essential safety practices and compliance standards to ensure efficient and reliable solar energy installations.

Elements and Performance Criteria

Mounting and Racking Systems

To be competent, the user/individual on the job must be able to:

- PC1.** Identify and select appropriate mounting and racking options for both roof-mounted and ground-mounted solar PV systems.
- PC2.** Install roof-mounted and ground-mounted systems effectively, considering structural requirements and obtaining necessary permits.

Solar Tracking Systems

To be competent, the user/individual on the job must be able to:

- PC3.** Identify and describe the types of solar tracking systems, including single-axis and dual-axis, and their respective operational mechanisms.
- PC4.** Evaluate the benefits and challenges of solar tracking systems in maximizing sunlight exposure and enhancing system efficiency.

PV Cells Selection and Sizing

To be competent, the user/individual on the job must be able to:

- PC5.** Evaluate solar cell characteristics, power output, and performance factors, including fill factor, STC, NOCT, and types of cells, especially silicon-based.

Solar Energy Storage Systems

To be competent, the user/individual on the job must be able to:

- PC6.** Select and implement battery technologies, DC-DC converters, charge controllers, and DC-AC inverters, including single-phase, three-phase, and MPPT, for effective solar energy storage systems.

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Inverters Selection and Sizing (Grid Connection and Off-Grid)

To be competent, the user/individual on the job must be able to:

PC7. Select and size inverters for both grid-connected and off-grid systems, understanding their purposes, types, interfaces, protection systems, and power quality monitoring.

Cable Selection and Sizing

To be competent, the user/individual on the job must be able to:

PC8. Accurately size AC and DC cables for solar installations, including from the inverter to ACDB, ACDB to MDB, and module to SMB to inverter.

Earthing Selection and Sizing

To be competent, the user/individual on the job must be able to:

PC9. Select and size appropriate earthing types, vertical electrodes, and the cross-sectional area of GI strips for effective grounding in solar installations.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Different mounting and racking systems for solar panels, including installation methods and structural requirements.
- KU2.** Types of solar tracking systems (single-axis and dual-axis) and their impact on maximizing sunlight exposure.
- KU3.** Characteristics, power output, and performance factors of PV cells, including fill factor, STC, NOCT, and different cell types.
- KU4.** Various solar energy storage systems, including battery technologies, DC-DC converters, charge controllers, and inverters (single-phase, three-phase, MPPT).
- KU5.** Criteria for selecting and sizing cables and earthing systems, including types of cables, earthing methods, and cross-sectional areas for effective grounding.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Design and configure solar PV systems, including selecting and installing appropriate mounting, racking, and tracking systems, while considering structural requirements and permitting.
- GS2.** Select and size PV cells, energy storage components, and system components such as batteries, DC-DC converters, charge controllers, and inverters to ensure optimal performance.
- GS3.** Assess and implement safety practices and compliance standards related to solar panel installations, including electrical safety and structural integrity for mounting systems.
- GS4.** Perform accurate cable and earthing selection and sizing for effective system operation and grounding, ensuring all components meet necessary specifications and standards.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Mounting and Racking Systems</i>	50	23	-	-
PC1. Identify and select appropriate mounting and racking options for both roof-mounted and ground-mounted solar PV systems.	-	-	-	-
PC2. Install roof-mounted and ground-mounted systems effectively, considering structural requirements and obtaining necessary permits.	-	-	-	-
<i>Solar Tracking Systems</i>	-	-	-	-
PC3. Identify and describe the types of solar tracking systems, including single-axis and dual-axis, and their respective operational mechanisms.	-	-	-	-
PC4. Evaluate the benefits and challenges of solar tracking systems in maximizing sunlight exposure and enhancing system efficiency.	-	-	-	-
<i>PV Cells Selection and Sizing</i>	-	-	-	-
PC5. Evaluate solar cell characteristics, power output, and performance factors, including fill factor, STC, NOCT, and types of cells, especially silicon-based.	-	-	-	-
<i>Solar Energy Storage Systems</i>	-	-	-	-
PC6. Select and implement battery technologies, DC-DC converters, charge controllers, and DC-AC inverters, including single-phase, three-phase, and MPPT, for effective solar energy storage systems.	-	-	-	-
<i>Inverters Selection and Sizing (Grid Connection and Off-Grid)</i>	-	-	-	-
PC7. Select and size inverters for both grid-connected and off-grid systems, understanding their purposes, types, interfaces, protection systems, and power quality monitoring.	-	-	-	-
<i>Cable Selection and Sizing</i>	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC8. Accurately size AC and DC cables for solar installations, including from the inverter to ACDB, ACDB to MDB, and module to SMB to inverter.	-	-	-	-
<i>Earthing Selection and Sizing</i>	-	-	-	-
PC9. Select and size appropriate earthing types, vertical electrodes, and the cross-sectional area of GI strips for effective grounding in solar installations.	-	-	-	-
NOS Total	50	23	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0912
NOS Name	Solar Panel Installation
Sector	Electronics
Sub-Sector	
Occupation	Electronics Assembly, Electronics Equipment Services
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

NIE/ELE/N0814: Safety, Maintenance, and Repairs of Solar Panel

Description

Safety, Maintenance, and Repairs of Solar Panel equips learners with skills for preventive and corrective maintenance of solar panels, including failure analysis and tool usage. It covers safety protocols for high-voltage systems, compliance with electrical codes, and effective commissioning, testing, and troubleshooting of solar energy systems, including battery maintenance.

Scope

The scope covers the following :

- Safety, Maintenance, and Repairs of Solar Panel covers preventive and corrective maintenance of solar panels, including troubleshooting and repair. It includes system commissioning, performance testing, and battery maintenance. Learners will also master safety protocols for high-voltage systems, adhere to electrical codes, and manage fire safety and emergency procedures.

Elements and Performance Criteria

Commissioning, Testing, and Maintenance

To be competent, the user/individual on the job must be able to:

- PC1.** Perform preventive and corrective maintenance on solar panels.
- PC2.** Conduct system commissioning and performance testing accurately.
- PC3.** Implement procedures for panel cleaning and vegetation management.

Safety, Codes, and Regulations

To be competent, the user/individual on the job must be able to:

- PC4.** Follow safety protocols for working with PV systems.
- PC5.** Apply high voltage electrical safety practices.
- PC6.** Adhere to national and local electrical codes and regulations.
- PC7.** Implement fire safety and emergency procedures effectively.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Basics of solar panel maintenance, including preventive and corrective maintenance.
- KU2.** Failure patterns of PV panel components and appropriate maintenance tools and equipment.
- KU3.** Procedures for system commissioning and performance testing to ensure optimal operation.
- KU4.** Safety protocols for working with high-voltage PV systems and implementing safety measures.
- KU5.** National and local electrical codes, regulations, fire safety, and emergency procedures.

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** Perform preventive and corrective maintenance on solar panels and associated components.
- GS2.** Implement safety protocols and practices when working with high-voltage PV systems.
- GS3.** Adhere to national and local electrical codes and regulations for compliance.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Commissioning, Testing, and Maintenance</i>	50	22	-	-
PC1. Perform preventive and corrective maintenance on solar panels.	-	-	-	-
PC2. Conduct system commissioning and performance testing accurately.	-	-	-	-
PC3. Implement procedures for panel cleaning and vegetation management.	-	-	-	-
<i>Safety, Codes, and Regulations</i>	-	-	-	-
PC4. Follow safety protocols for working with PV systems.	-	-	-	-
PC5. Apply high voltage electrical safety practices.	-	-	-	-
PC6. Adhere to national and local electrical codes and regulations.	-	-	-	-
PC7. Implement fire safety and emergency procedures effectively.	-	-	-	-
NOS Total	50	22	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0814
NOS Name	Safety, Maintenance, and Repairs of Solar Panel
Sector	Electronics
Sub-Sector	
Occupation	Electronics Equipment Services, Electronics Assembly
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

NIE/ELE/N0922: Implementation of Solar Power Electronics Project

Description

Nature: The Certificate course is targeted for creating qualified professionals in the field of Solar Panel Installation. Qualification has been developed in consultation with industry experts in the domain, aiming at Empowering the future workforce with necessary skills for employment and entrepreneur development of the qualifier. **Purpose:** The objective of this O Level (Solar Panel Installation and Maintenance Technician) course is to equip candidates with the knowledge and practical skills needed to safely and effectively install and maintain photovoltaic systems for renewable energy generation.

Scope

The scope covers the following :

- The Solar Panel Installation and Maintenance Technician course covers four key areas: solar technology fundamentals, electrical basics, installation techniques, and system safety and maintenance. Learners gain in-depth knowledge of solar energy, panel types, system configurations, and site assessment. The course also includes electrical principles, PV module connections, and inverter integration. Practical training focuses on mounting systems, battery storage, cable sizing, and troubleshooting. Emphasis is placed on preventive maintenance and safety protocols. The full program spans 480 hours, combining theory and hands-on experience to prepare students for real-world solar installations.

Elements and Performance Criteria

Install and Configure Solar PV Modules

To be competent, the user/individual on the job must be able to:

- PC1.** Select appropriate mounting systems based on rooftop or ground conditions.
- PC2.** Connect solar PV modules in series and parallel as per system design.
- PC3.** Ensure correct orientation, tilt, and spacing of panels to optimize energy output.

Assess Site and Solar Radiation

To be competent, the user/individual on the job must be able to:

- PC4.** Use instruments to measure global, beam, and diffuse solar radiation.
- PC5.** Conduct site and shade analysis using standard methods.
- PC6.** Identify suitable installation locations based on solar access and load requirements.

Maintain and Troubleshoot Solar PV Systems

To be competent, the user/individual on the job must be able to:

- PC7.** Perform routine maintenance including panel cleaning and vegetation control.
- PC8.** Use appropriate tools to diagnose faults in inverters, batteries, and connections.
- PC9.** Replace or repair faulty system components following safety procedures.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** Possess a clear understanding of solar energy concepts, system types, and the role of solar power in sustainable development.
- KU2.** Apply fundamental electrical knowledge and understand the working of key solar components like panels, inverters, and batteries.
- KU3.** Demonstrate practical skills in installing, maintaining, and troubleshooting solar PV systems using appropriate tools and techniques.
- KU4.** Follow workplace protocols, safety standards, and documentation procedures while coordinating effectively within the organization.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and interpret technical guidelines, communicate clearly with teams and customers, and maintain accurate reports on solar installations and maintenance.
- GS2.** Organize tasks efficiently, choose the right solar components, and manage resources effectively to ensure smooth installation and operation.
- GS3.** Diagnose and resolve system faults, optimize performance, and apply analytical thinking to enhance safety, efficiency, and customer satisfaction.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Install and Configure Solar PV Modules</i>	-	-	-	160
PC1. Select appropriate mounting systems based on rooftop or ground conditions.	-	-	-	-
PC2. Connect solar PV modules in series and parallel as per system design.	-	-	-	-
PC3. Ensure correct orientation, tilt, and spacing of panels to optimize energy output.	-	-	-	-
<i>Assess Site and Solar Radiation</i>	-	-	-	-
PC4. Use instruments to measure global, beam, and diffuse solar radiation.	-	-	-	-
PC5. Conduct site and shade analysis using standard methods.	-	-	-	-
PC6. Identify suitable installation locations based on solar access and load requirements.	-	-	-	-
<i>Maintain and Troubleshoot Solar PV Systems</i>	-	-	-	-
PC7. Perform routine maintenance including panel cleaning and vegetation control.	-	-	-	-
PC8. Use appropriate tools to diagnose faults in inverters, batteries, and connections.	-	-	-	-
PC9. Replace or repair faulty system components following safety procedures.	-	-	-	-
NOS Total	-	-	-	160

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0922
NOS Name	Implementation of Solar Power Electronics Project
Sector	Electronics
Sub-Sector	
Occupation	Electronics Assembly, Electronics Equipment Services
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

This section includes the processes involved in identifying, gathering, and interpreting information to evaluate the Candidate on the required competencies of the program.

Assessment of the qualification evaluates candidates to ascertain that they can integrate knowledge, skills and values for carrying out relevant tasks as per the defined learning outcomes and assessment criteria. The underlying principle of assessment is fairness and transparency. The evidence of the outcomes and assessment criteria. Competence acquired by the candidate can be obtained by conducting Theory (Online), Practical assessment, internal assessment, Project/Presentation/ Assignment, Major Project. The emphasis is on the practical demonstration of skills & knowledge gained by the candidate through the training. Each OUTCOME is assessed & marked separately. A candidate is required to pass all OUTCOMES individually based on the passing criteria.

About Examination Pattern:

1. The question papers for the theory exams are set by the Examination wing (assessor) of NIELIT HQS.
2. The assessor assigns the roll number.

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3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.
4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.
5. Pass percentage would be 50% marks in each component.
6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).
7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.
8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.
9. The examination will be conducted in English language only.

Quality assurance activities: A pool of questions is created by a subject matter expert and moderated by other SME. Test rules are set beforehand. Random set of questions which are according to syllabus appears which may differ from candidate to candidate. Confidentiality and impartiality are maintained during all the examination and evaluation processes.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ELE/N0910.Solar Technology Fundamentals	50	22	-	-	72	14
NIE/ELE/N0911.Electric Fundamentals for Solar System	50	23	-	-	73	15
NIE/ELE/N0912.Solar Panel Installation	50	23	-	-	73	15
NIE/ELE/N0814.Safety, Maintenance, and Repairs of Solar Panel	50	22	-	-	72	14
NIE/ELE/N0922.Implementation of Solar Power Electronics Project	-	-	-	160	160	32
Total	220	120	-	160	500	100

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.